Stat 537

Homework 7

October 5, 2026

1. For a 2 treatment, 3 block, randomized block design with no replication, (i.e., there are six observations), write out **Y, X, $\boldsymbol{\beta}$, Z, u,** and **e**.

2. Generate random data for a two treatment three block randomized block design. It will have ten blocks and 5 observations per treatment per block, for a total of 100 observations. Let the treatment means be 100 and 110, let $\sigma_e^2 = 16$ and $\sigma_b^2 = 12$.

3. Run the data for the previous problem through `Proc Mixed`. SAS will use REML as the default. How well do the estimates of the analysis reflect the actual values used to generate the data? Both fixed effects and variance components.

4. What is the value of the F-statistic to test the null hypothesis of no treatment effect?

5. Use R to analyze the same data using REML. Compare your answers.

6. Now analyze the data in R using maximum likelihood instead of REML. How the estimates for the variance components change.

7. Perform a likelihood ratio test of the null hypothesis that $\sigma_b^2 = 0$, using both REML and ML likelihoods. What is the p-value? Did you remember to cut it in half?

8. Perform a likelihood ratio test of the null hypothesis of no treatment difference. What are the degrees of freedom? What is the p-value? How does it compare to the F-test?

9. Now randomly remove 20% of the data. Use `Proc Mixed` and `ddfm=Satterthwaite`. How do the degrees of freedom change in the F-test?

10. How do the degrees of freedom change in the estimates of the treatment means?

11. Now use the option `ddfm=KenwardRoger`. Comment on changes you see in the output.

12. Now add a random effect for `treatment*block` in the analysis even though you know one does not exist in the data. Perform a likelihood ratio test on the null hypothesis $\sigma_{tb}^2 = 0$. Do you reject the null?